Group 193 Zoo Lab

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2) Description

An immersive audio track about axolotls, with fun facts and explanations of why they are critically endangered. The Toronto Zoo would have a poster with the QR code, allowing visitors to listen while observing. Simple, yet detailed language allows for an engaging experience for both the target group of Grade 6 students and all other visitors.

3) Sharing

We would like to share our project with Toronto Zoo.

4) Summary

Axolotls can't control their body temperature and climate change makes their habitat warmer, making it more hostile to live in. However, conservation efforts led by the University of Mexico and local Mexican farmers demonstrate that science and culture can work together to protect axolotl habitats.

5a) Scientific Explanation

The vulnerability of axolotls is due to their lack of thermoregulation as they are part of the amphibian class. Since they are endothermic, changes in water conditions from climate change are affecting their physiological processes. Because of the increase in water temperature, egg and testis production has decreased. Additionally, from the growth and development perspective the rising water temperatures mean eggs having difficulties hatching and smaller mature body size of the few that do survive (Figiel, 2023).

Sustainability efforts utilize both modern science and local farming, as researchers from the University of Mexico are working with local farmers to create chinampa-refuges near Xochimilco. These sanctuaries utilize man-made farming islands constructed during the Aztec era, along with modern biofiltration systems. This results in high-quality water conditions and limits the presence of invasive predators, providing a safer habitat for many aquatic animals, especially Axolotls (McCoy, 2025).

Population data shows that Axolotls are officially classified as Critically Endangered. Currently, the number of mature axolotls in the wild range from 50 to 1,000 individuals and numbers are continuing to decline. Their natural freshwater wetland habitat in central-southern Mexico is shrinking and becoming increasingly uninhabitable due to climate change and pollution, further threatening their survival (Diaz et al., 2020).

5b) References

Diaz, O. L., Buendia, A., Sánchez, J., Villalobos, G., Rojas-Serrania, N., Cervantes, J. A. O., ... & Muñoz-García, C. I. (2025). Identification of enterobacteriaceae causing septicemia in the axolotl Ambystoma mexicanum. *Antonie van Leeuwenhoek*, *118* (3), 56.https://doi.org/10.2305/IUCN.UK.2020-3.RLTS.T1095A53947343.en

Figiel C. R., Jr (2023). Effects of Water Temperature on Gonads Growth in *Ambystoma mexicanum* Axolotl Salamanders. *Animals : an open access journal from MDPI*, *13* (5), 874. https://doi.org/10.3390/ani13050874

McCoy, M. K. (2025). *Axolotls are on the brink: Can we bring them back?* Conservation International. https://www.conservation.org/news/axolotls-are-on-the-brink-can-we-bring-them-back

6) Submission

https://publications.rosemax.uk/zoo193.mp3

i) Transcript

These are Axolotls. Yes, the same ones from Minecraft. They're fascinating species of salamanders with remarkable abilities, including staying in a young form for life, having a variety of colours, and the ability to regrow lost limbs.

As incredible as they may seem, these cute creatures are critically endangered.

In the past 20 years, wild populations have decreased 99% in wetlands, even in their native lake of Xochimilco, Mexico and only an estimated 500 are left in the wild.

Like other amphibians, they are cold-blooded, meaning they cannot perform thermoregulation. Their environment controls their body temperatures, making them vulnerable to harmful changes.

With climate change, the water temperatures have reached an average of 24°C; 5° higher than their ideal conditions. In turn, their reproduction has been affected, with their eggs having trouble hatching and their bodies maturing smaller.

But not everything about these little guys is hostile. Many people are taking action right now to help axolotls to survive for the future.

Researchers from the University of Mexico are partnering with local farmers, creating chinampa-refuges, which are Aztec-era farm islands, and using special biofilters made of natural materials to improve water quality and block invasive predators.

As we continue with our efforts of their conservation, the future of our favourite amphibians could look promising. Think about what you could do to help them!